

1.2 Lengths / Angles from Dot Products.

向量的形式

V. 的矩阵形式:

$$\begin{bmatrix} v_1 & v_2 \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = v_1 w_1 + v_2 w_2$$

1. vector multiplication

→ one way: dot product

$$V = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$V \cdot W = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$$

$$\Rightarrow V \cdot V = \|V\|^2 = v_1^2 + v_2^2 + \dots + v_n^2 : \text{the squared length of a vector.}$$

$$(2) \text{ unit vector: } \frac{V}{\|V\|} = u$$

$$(3) \text{ perpendicular vectors: } V \cdot W = 0$$

$$\rightarrow \text{e.g.: } V = (1, 0) \quad u = (\cos \theta, \sin \theta)$$

$$\Rightarrow V \cdot u = \cos \theta$$

$$\Rightarrow \text{引出 } \frac{V}{\|V\|} \cdot \frac{W}{\|W\|} = \cos \theta$$

$$\Rightarrow V \cdot W = \|V\| \cdot \|W\| \cdot \cos \theta = \text{the definition of dot product}$$

↓
angle of —

2.

SCHWARZ INEQUALITY

$$|v \cdot w| \leq \|v\| \|w\|$$

TRIANGLE INEQUALITY

$$\|v + w\| \leq \|v\| + \|w\|$$

ality comes directly from the Schwarz inequality!